The subexponential factor of meander numbers

Meander numbers: growth rate and exponent. Paper C.

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Abstract

A meander of order n is a closed road that crosses a straight river $2n$ times without crossing itself. Their number M_n grows like μ^n with $\mu \approx 12.26$, and the factor $u_n = M_n/\mu^n$ that remains is conjectured to decay like a power $n^{-\alpha}$ with $\alpha = (29+\sqrt{145})/12 \approx 3.4201$. This paper collects what can be proved about u_n. The main result is that $\mu^n e^{-C\sqrt{n}} \le M_n \le \mu^{n+1}/(8n)$ for every n. The lower bound is new; it removes a logarithmic factor from the previous one. Its proof cuts a meander into two balanced halves and completes each half to a meander in a way that can be undone at small cost. Around these bounds we prove the ratio limit $M_{n+1}/M_n \to \mu$ with a rate, the unique factorization of meanders into prime meanders (those that are not obtained by gluing two smaller ones), the bound $\sum_n u_n \le 1/4$, a bound on the probability that a meander splits as such a product at a given position, and comparisons with splittings that are nearly exact. For meanders whose crossing labels, a height attached to each crossing, stay in $\{-1,0,1,2\}$, the growth rate and the exponent are found exactly: the growth rate is a quartic irrational $10.4245\ldots$ and the exponent is $3.7758\ldots$, given by the angle of a wedge in which a two-dimensional walk is confined. The generating function of this bounded-label family is not D-finite: it satisfies no linear differential equation with polynomial coefficients. Finally we state which estimates would improve the bounds, show why some natural routes cannot, and list exact data. All code was written using Claude (Opus 5.5) or Codex (Sol 5.1) with active review and guidance throughout.

1 Introduction

1.1 The factor u_n

A *meander of order* n is a closed road that crosses a straight river $2n$ times and never crosses itself. Number the crossings $1,\ldots,2n$ from left to right along the river. Above the river the road consists of n *arches*, each joining two crossings, and no two arches cross; the same holds below the river. So a meander is a pair of noncrossing perfect matchings of $\{1,\ldots,2n\}$, one drawn above and one below, whose union is a single closed curve. Figure 1 shows one of the eight meanders of order 3. The number M_n of meanders of order n starts

$$1,\ 2,\ 8,\ 42,\ 262,\ 1828,\ 13820,\ 110954,\ \ldots$$

and is known exactly for every $n \le 32$ (OEIS A005315 [20] and [13, 14]). No formula is known.

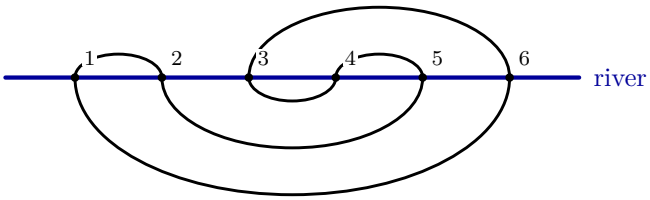


Figure 1: A meander of order 3, the running example of the series. Above the river the road joins crossings 1–2, 3–6 and 4–5; below it joins 1–6, 2–5 and 3–4.

The growth rate $\mu = \lim_n M_n^{1/n}$ exists (Section 2). Its value is about 12.26, and Paper B of this series proves $12.2191 \le \mu \le 12.2970$ [22]. This paper is about what is left once the exponential growth is removed,

$$u_n \;=\; \frac{M_n}{\mu^n}.$$

Intuitively, μ^n accounts for the local freedom of the road, about μ choices for each new pair of crossings, and u_n is the remaining global correction. The physics prediction [6] is

$$u_n \;\approx\; A\, n^{-\alpha}, \qquad \alpha = \alpha_* = \frac{29+\sqrt{145}}{12} \approx 3.4201, \tag{1}$$

for a constant A. The prediction is not proved; numerical analysis of the exact counts supports it [15]. Paper A explains where the number comes from [23], and Paper D studies the continuum model behind it [21]. Throughout, α denotes the true exponent, if it exists, and α_* the predicted number. We use the logarithmic form

$$L(n) \;=\; n\log\mu - \log M_n \;=\; -\log u_n,$$

so that (1) reads $L(n) = \alpha\log n + O(1)$. We call $L(n)$ the *loss* at order n.

1.2 Main results

Rigorous work on meanders has concentrated on the growth rate [1, 22]. We are not aware of earlier rigorous bounds on u_n beyond $u_n \le 1$, which follows from the existence of μ; our literature search was limited, and we claim no priority. The law (1) could fail in two ways: u_n could decay faster than every power, or it could be irregular. The results below address both.

Theorem 1. *There is a constant C such that for every $n \ge 1$*

$$\mu^n e^{-C\sqrt{n}} \;\le\; M_n \;\le\; \frac{\mu^{n+1}}{8n}.$$

So u_n lies between $e^{-C\sqrt{n}}$ and $\mu/(8n)$. The upper bound on u_n is already a power law, with exponent 1 in place of 3.42; the lower bound $e^{-C\sqrt{n}}$ decays faster than every power. In logarithmic terms,

$$\log n - \log(\mu/8) \;\le\; L(n) \;\le\; C\sqrt{n},$$

while the prediction is $L(n) = 3.4201\log n + O(1)$. The upper bound on M_n comes from gluing two meanders of order n in at least $4n-1$ ways that can be told apart afterwards (Section 3).

The lower bound is new. It improves the bound $L(n) \le C'\sqrt{n}\log n$ obtained earlier in this project, and it comes from cutting a meander into two balanced pieces along a closed curve that meets the road and the river about $\sqrt{n}$ times (Section 4).

Around Theorem 1 we prove or collect the following.

- *Growth and ratios* (Section 2). The ratio limit $M_{n+1}/M_n \to \mu$, with the rate $|M_{n+1}/(\mu M_n) - 1| \le C_{\mathrm{r}}\, n^{-1/4}\sqrt{\log n}$ for a constant C_{r}, improved to $O(n^{-1/4})$ by the lower bound of Theorem 1, and an exact *halving identity* that expresses $L(n)$ through the numbers $D(m) = \log(M_{2m}/M_m^2)$, which measure how much more common a meander of order $2m$ is than a pair of meanders of order m.

- *A subsequence result* (Section 3). For every $\kappa < 3/2$, $\liminf_n n^{\kappa} u_n = 0$. So if a power law holds, then $\alpha \ge 3/2$.

- *Products and cuts* (Section 5). Two meanders can be glued side by side into one, their *product*; a *cut* is a position where a meander comes apart as such a product. Every meander factors uniquely into *prime* meanders, those without a cut; $\sum_n u_n \le 1/4$; the expected number of cuts is bounded; the probability of a cut at distance s from the ends is at most $C_{\mathrm{c}}\, s^{-1/4}\sqrt{\log(s+1)}$; and *near-cuts*, which fail to be cuts in at most K crossings, are at most $e^{O(K+1)}$ times more frequent than cuts.

- *An exactly solvable family* (Section 6). Each crossing of a meander carries an integer *label*, a signed count of the half turns the road makes on its way from the first crossing. For meanders whose labels lie in $\{-1, 0, 1, 2\}$, the count is $\Lambda^n n^{-\alpha_{\mathrm{b}}+o(1)}$ with $\Lambda = 10.4245\ldots$ and $\alpha_{\mathrm{b}} = 3.7758\ldots$, both in closed form. Because α_{b} is irrational, the generating function of this family is not D-finite: it satisfies no linear differential equation with polynomial coefficients.

- *Criteria* (Section 7). Two conditional statements that would improve the lower bound on M_n to $L(n) = O(n^{\tau})$ with $\tau < 1/2$, a proof that gluing short meanders cannot supply their hypotheses, a heuristic for what is true, and an exact reformulation of a weaker target.

1.3 Status of the results

Besides the lower bound of Theorem 1, the following are new as far as we know: the upper bound, the subsequence result, the bound $\sum u_n \le 1/4$, the cut bounds, the near-cut comparisons, the results on bounded labels, including the corollary that the generating function of that family is not D-finite (its main input is the exponent theorem), and the criteria of Section 7. The factorization into prime meanders is a close relative of known decompositions, and we claim no novelty for it.

Most results were checked by a separate, independent reading of the written proof, which found no gap. This includes both bounds of Theorem 1, the subsequence result, the ratio limit and its rate (also in the improved form that uses Theorem 1), the factorization, $\sum u_n \le 1/4$, the near-cut comparisons, the growth rate and the exponent for bounded labels, the irrationality of the exponent α_{b}, and the corollary on D-finiteness. The halving identity and its corollary, the transcendence of α_{b}, the cut bound of Proposition 16 and the criteria of Section 7 have complete written proofs that have not been checked in this way. Appendix B gives the status

of each result. Nothing in this paper has been reviewed outside the project. Conjectures, heuristics and numerical statements are labeled as such. The exact data in Appendix A use only the published counts M_n, $n \le 32$.

2 Gluing and the growth rate

2.1 Terminology

Let a meander of order N have crossings $1, \dots, 2N$. The *gap* i, for $0 \le i \le 2N$, is the stretch of river between crossings i and $i+1$; gaps 0 and $2N$ are the two ends. An arch joining crossings $x < y$ *passes over* gap i if $x \le i < y$. An arch is *exterior* if no other arch on the same side encloses it, and *short* if it joins two neighboring crossings. The probability $\Pr_N$ and the expectation $\mathbb{E}_N$ refer to a meander of order N chosen uniformly at random.

2.2 Supermultiplicativity and Fekete's lemma

Two meanders can be glued into one. Let G have order a and H order b. Place H to the right of G on the same river. The last crossing $2a$ of G is the right end of an exterior upper arch $\{x, 2a\}$, and the first crossing of H, now numbered $2a+1$, is the left end of an exterior upper arch $\{2a+1, y\}$. Replace these two arches by $\{x, y\}$ and $\{2a, 2a+1\}$. Nothing crosses, because $\{x, y\}$ passes over everything between x and y; and re-pairing one arch of each of two closed curves joins them into one. Intuitively, both roads are cut open next to the boundary between them and reconnected across it. The result $G \cdot H$ is a meander of order $a+b$, the *product* of G and H. Given a, the pieces are recovered by undoing the re-pairing. Hence

$$M_{a+b} \;\ge\; M_a M_b \qquad \text{for all } a, b \ge 1. \tag{2}$$

Section 5 identifies exactly which meanders are products.

Inequality (2) says that $\log M_n$ is superadditive. Fekete's lemma [8] then says that $\log M_n / n$ converges to its supremum. The limit is finite because the upper and lower arch systems are each counted by the Catalan number $\mathrm{Cat}_n = \binom{2n}{n}/(n+1)$, so $M_n \le \mathrm{Cat}_n^2 \le 16^n$. Hence $\mu = \lim M_n^{1/n}$ exists, $\mu \le 16$, and

$$M_n \;\le\; \mu^n, \qquad \text{that is} \qquad u_n \le 1 \quad \text{and} \quad L(n) \ge 0 \qquad \text{for every } n.$$

In terms of L, inequality (2) says that L is *subadditive*,

$$L(a+b) \;\le\; L(a) + L(b), \tag{3}$$

and the existence of μ says $L(n)/n \to 0$. These three facts about L (it is nonnegative, subadditive, and $o(n)$) are used repeatedly.

2.3 The ratio limit

Theorem 2 (ratio limit)**.** *$M_{n+1}/M_n \to \mu$. In particular the largest ratio $R_\star = \sup_n M_{n+1}/M_n$ is finite.*

The proof rests on an exact identity. A *wiggle* is a short upper arch followed by a short lower arch, on three consecutive crossings $i, i+1, i+2$. Any crossing x of a meander of order n can be replaced by three consecutive crossings that form a wiggle, with the old arches at x reattached to the outer two. This keeps the road one loop and gives a meander of order $n+1$ with a marked wiggle; collapsing the marked wiggle undoes it. So marked crossings of order-n meanders correspond exactly to marked wiggles of order-$(n+1)$ meanders, and with Z the number of wiggles,

$$M_{n+1}\,\mathbb{E}_{n+1}[Z] \;=\; 2n\,M_n.$$

A further counting argument with these insertions shows that Z/n concentrates at $2/\mu$ with exponentially small error probabilities, in both directions. The identity then gives $M_{n+1}/M_n = 2n/\mathbb{E}_{n+1}[Z] \to \mu$. Since $Z \le n+1$, it also gives $M_{n+1} \ge M_n$: the counts never decrease.

The ratio limit has a rate. Its proof combines a lower bound on M_n with two inequalities that hold for every $n \ge 3$, with a constant C_0:

$$\frac{M_{n+1}}{\mu M_n} \;\ge\; 1 - C_0\sqrt{\frac{L(n)+\log n}{n}}, \qquad \frac{\mu M_n}{M_{n+1}} \;\ge\; 1 - C_0\sqrt{\frac{L(n+1)+\log n}{n}}. \tag{4}$$

The first bounds the ratio from below and the second bounds it from above. Both come from a further structural decomposition of meanders that we do not describe here [18]. A lower bound on M_n is an upper bound on the loss L, and it makes both right sides close to 1.

Theorem 3 (rate in the ratio limit)**.** *There is a constant C_{r} such that for every $n \ge 2$*

$$\left|\frac{M_{n+1}}{\mu M_n} - 1\right| \;\le\; C_{\mathrm{r}}\, n^{-1/4}\sqrt{\log n}.$$

Consequently $|\log(u_{n+k}/u_n)| \le C'_{\mathrm{r}}\, k\, n^{-1/4}\sqrt{\log n}$ for $0 \le k \le n$.

The constants are large: with the constants of the proof the bound says something only for n beyond about 10^{14}. The proof inserts the earlier lower bound $L(n) \le C'\sqrt{n}\log n$ into (4). With Theorem 1 in its place the factor $\sqrt{\log n}$ disappears, by the following short argument, which we give in full.

Remark. There is a constant C''_{r} such that $|M_{n+1}/(\mu M_n) - 1| \le C''_{\mathrm{r}}\, n^{-1/4}$ for every $n \ge 2$. Indeed, by Theorem 1, $L(n) \le C\sqrt{n}$ and $L(n+1) \le 2C\sqrt{n}$, so both right sides of (4) are at least $1-\epsilon_n$ with $\epsilon_n = O(n^{-1/4})$; when $\epsilon_n \le 1/2$ the second inequality gives $M_{n+1}/(\mu M_n) \le 1/(1-\epsilon_n) \le 1+2\epsilon_n$, and the finitely many other n are covered by $M_{n+1}/M_n \le R_\star$ after enlarging the constant. Summing the logarithms as in Theorem 3 gives $|\log(u_{n+k}/u_n)| = O(k\,n^{-1/4})$ for $0 \le k \le n$, and the proof of Proposition 16 below, run with this rate, improves the cut bound to $O(s^{-1/4})$. The inequalities (4) cannot give more than $n^{-1/4}$: sequences of the same form as the decomposition behind them, with loss of order $\sqrt{n}$, can have ratios that differ from μ by a multiple of $n^{-1/4}$ infinitely often [18].

Theorem 3 says that u_n is *locally smooth*: it changes by a bounded factor over windows of length about $n^{1/4}/\sqrt{\log n}$, and by the remark over windows of length about $n^{1/4}$. It does not say that u_n is decreasing: the statement $M_{n+1} \le \mu M_n$ for all n is open. It holds for every $n \le 31$, since the ratios M_{n+1}/M_n increase in that range and $M_{32}/M_{31} = 11.023$ is below the lower bound 12.2191 for μ.

2.4 The halving identity

The *doubling defect* is

$$D(m) \;=\; \log \frac{M_{2m}}{M_m^2} \;=\; 2L(m) - L(2m).$$

By (2), $D(m) \geq 0$. Section 5 shows that $e^{-D(m)}$ is exactly the probability that a random meander of order $2m$ is a product of two meanders of order m.

Proposition 4 (halving identity)**.** *For every $n \geq 1$,*

$$L(n) \;=\; \sum_{j \geq 0} 2^{-(j+1)} D(2^j n).$$

Proof. Since $D(m) = 2L(m) - L(2m)$, the partial sums telescope: $\sum_{j=0}^{J-1} 2^{-(j+1)} D(2^j n) = L(n) - 2^{-J} L(2^J n)$. The last term is $n \cdot L(2^J n)/(2^J n)$, which tends to 0 because $L(k)/k \to 0$. □

The right side involves only the counts, not μ. Intuitively, the identity starts from a huge meander, halves it again and again down to order n, and adds up what each halving costs. It turns statements about u_n into statements about the single sequence $D(m)$.

Corollary 5. *For a constant α:*

1. *$L(n) = \alpha \log n + O(1)$ for all n if and only if $D(m) = \alpha \log m + O(1)$;*
2. *$L(n) = \alpha \log n + o(\log n)$ if and only if $D(m) = \alpha \log m + o(\log m)$;*
3. *$L(n) \leq \chi \log n + O(1)$ for some χ if and only if $D(m) \leq \chi' \log m + O(1)$ for some χ'.*

Proof. In each part, one direction follows from $D(m) = 2L(m) - L(2m)$ and the other from the identity. For example, in part 1, if $|D(m) - \alpha \log m| \leq c$ for all m, then

$$L(n) = \sum_{j \geq 0} 2^{-(j+1)} \big(\alpha \log n + \alpha j \log 2 + O(c)\big) = \alpha \log n + \alpha \log 2 + O(c),$$

using $\sum_j 2^{-(j+1)} = 1$ and $\sum_j j\, 2^{-(j+1)} = 1$. Part 3 is the same computation with inequalities, using $L \geq 0$, and part 2 uses that $\epsilon \log(2^j n) \leq \epsilon(\log n + j \log 2)$. □

So the power law (1) is equivalent to $D(m) = \alpha \log m + O(1)$: the probability that a meander of order $2m$ splits into two meanders of order m should decay like $m^{-\alpha}$. If $u_n \sim A n^{-\alpha}$, then $D(2m) - D(m)$ tends to $\alpha \log 2 \approx 2.371$; if in addition the correction is smooth, for example $u_n = A n^{-\alpha}(1 + O(1/n))$, then $m\,(D(m+1) - D(m))$ tends to α. Table 1 in Appendix A lists exact values. Along the computed range D is increasing and concave, and both of these quantities increase, reaching 3.13 at $m = 15$ and 2.19 at $m = 8$. Theorem 1 gives $D(m) \leq 2L(m) \leq 2C\sqrt{m}$.

3 The upper bound

The bound $M_n \leq \mu^n$ uses one way of gluing two meanders. The bound $M_n \leq \mu^{n+1}/(8n)$ uses many: if two meanders of order n can be glued in at least $4n - 1$ ways that can be told apart afterwards, then meanders of order about $2n$ are about n times as many as pairs of meanders of order n, and that forces a factor $1/n$. The device of joining two objects of equal size in many recognizable ways goes back to work of Madras on lattice trees [17].

3.1 Capped blocks

Let Γ be a meander of order N. A *capped block* of Γ is a lower arch $\{\ell, \ell'\}$ with $\ell' \geq \ell + 3$ such that every upper arch with an end in $\{\ell + 2, \ldots, \ell'\}$ has both ends there. Its *size* is $\ell' - \ell + 1$, the number of crossings from ℓ to ℓ'. Intuitively, a capped block is a small meander hanging under one lower arch, attached to the rest of the road only through the two crossings ℓ and $\ell + 1$. The size is always even, $2b + 2$ with $b \geq 1$, and the road inside the block is a single path through all its crossings. Two lower arches are nested or disjoint, so two capped blocks are nested or disjoint. Two capped blocks with more than N crossings each, out of $2N$, must overlap and are therefore nested; so a meander has at most one capped block of each size larger than N.

Insertion. The upper arches of a meander cut the half-plane above the river into *regions*. Let G be a meander of order a, let i be one of its $2a + 1$ gaps, and let $\{p, q\}$ be an upper arch on the boundary of the region that lies directly above gap i. Let H be a meander of order b. Put $2b + 2$ new crossings into gap i: the crossings of H, with one extra crossing on each side, ℓ on the left and ℓ' on the right, joined by a new lower arch that caps them. Remove $\{p, q\}$ and the upper arch $\{\ell + 1, y_1\}$ at the first crossing of H. Add the upper arch $\{y_1, \ell'\}$, and join ℓ and $\ell + 1$ to p and q by two upper arches, in the one of the two possible pairings that crosses nothing. Because $\{p, q\}$ lies on the boundary of the region containing the gap, no other arch is crossed. The road becomes one loop, and the new lower arch spans a capped block of size $2b + 2$. Extraction reverses this. Intuitively, H is opened at its first crossing and spliced into the road of G at the arch $\{p, q\}$.

Theorem 6 (insertion bijection)**.** *For $a, b \geq 1$ and $N = a + b + 1$, insertion is a bijection from the quadruples $(G, i, \{p, q\}, H)$ onto the pairs (meander Γ of order N, capped block of Γ of size $2b + 2$).*

Write $T(G)$ for the number of pairs $(i, \{p, q\})$ available in G, and $T_a = \sum_G T(G)$, the sum over meanders G of order a. The theorem says that the meanders of order N, counted with their number of capped blocks of size $2b+2$, total $T_a M_b$. For $a \leq b$ the block has $2b+2 \geq N+1$ crossings and is unique, which gives

$$M_{a+b+1} \;\geq\; T_a M_b \qquad (1 \leq a \leq b). \tag{5}$$

Lemma 7. *For every meander G of order a, $T(G) \geq 4a - 1$. Hence $T_a \geq (4a - 1)M_a$.*

Proof. The a upper arches cut the upper half-plane into $a + 1$ regions; the pairs $(i, \{p, q\})$ counted by $T(G)$ are, region by region, a gap in it times an arch on its boundary. A bounded region lies directly under one arch with k arches directly below it; it has $k + 1$ arches on its boundary and contains $k+1$ gaps. The unbounded region has c exterior arches on its boundary and contains $c + 1$ gaps. So $T(G) = \sum_F c_F^2 + c_\infty$, where $c_F \geq 1$ is the number of arches on the boundary of region F and $c_\infty = c$. Every arch bounds two regions, so $\sum_F c_F = 2a$. For positive integers $c^2 \geq 3c - 2$, so $\sum_F c_F^2 \geq 6a - 2(a + 1) = 4a - 2$, and $c_\infty \geq 1$. □

3.2 The doubling argument

Proof of the upper bound in Theorem 1. Put $g(n) = 8n\, M_n$. By (5) with $a = b = n$ and Lemma 7,

$$g(2n + 1) = 8(2n + 1)M_{2n+1} \;\geq\; 8(2n + 1)(4n - 1)M_n^2 \;\geq\; 64n^2 M_n^2 = g(n)^2,$$

because $8(2n+1)(4n-1) = 64n^2+16n-8 \geq 64n^2$. Since $(2n+1)+1 = 2(n+1)$, the numbers $f(n) = \log g(n)/(n+1)$ satisfy $f(2n+1) \geq f(n)$. Let $n^{(0)} = n$ and $n^{(k+1)} = 2n^{(k)}+1$, so $n^{(k)}+1 = 2^k(n+1)$. Then

$$f(n) \;\leq\; f(n^{(k)}) \;=\; \frac{\log(8n^{(k)})}{n^{(k)}+1} + \frac{n^{(k)}}{n^{(k)}+1} \cdot \frac{\log M_{n^{(k)}}}{n^{(k)}} \;\longrightarrow\; \log\mu \qquad (k \to \infty).$$

So $\log g(n) \leq (n+1)\log\mu$, that is $8nM_n \leq \mu^{n+1}$. □

Every n starts its own chain $n, 2n+1, 4n+3, \ldots$, so the bound holds at every n and needs no regularity of u_n. The constant 8 cannot be improved by this argument. On the exact data, $(8nM_n)^{1/(n+1)} \leq 9.07$ for $n \leq 32$, well below μ, so the bound is far from tight there. In terms of the loss it says $L(n) \geq \log n - \log(\mu/8)$. Combined with $M_{2m+1} \leq R_\star M_{2m}$, inequality (5) with $a = b = m$ also gives $D(m) \geq \log(4m-1) - \log R_\star$; under the power law this says again that $\alpha \geq 1$.

3.3 The exponent $3/2$ along a subsequence

Insertion can be repeated. Let H_1 and H_2 be meanders of orders b_1 and b_2, and let $\mathrm{wrap}(H_2)$ be the meander of order b_2+1 in which H_2 hangs as a capped block under one new lower arch. Inserting the product $H_1 \cdot \mathrm{wrap}(H_2)$ into G gives a map that is again one-to-one with a recognizable image: the inserted block is the smallest capped block with more than N crossings. The result is a three-piece inequality.

Theorem 8 (three-piece insertion)**.** *For $N \geq 5$,*

$$M_N \;\geq\; \sum T_a\, M_{b_1} M_{b_2},$$

the sum over positive integers with $a+b_1+b_2 = N-2$, $2a \leq N-1$, $2b_1 \leq N$ and $2b_2+2 \leq N$.

Theorem 9. *For every $\kappa < 3/2$, $\liminf_{n\to\infty} n^\kappa u_n = 0$. Equivalently, $\limsup_n L(n)/\log n \geq 3/2$.*

Proof. Suppose $u_n \geq c_\kappa n^{-\kappa}$ for all large n, with $\kappa < 3/2$. If $\kappa < 1$ this already contradicts $u_n \leq \mu/(8n)$, so let $1 \leq \kappa < 3/2$. In Theorem 8 keep the triples with a and b_1 in $[N/4, 3N/8]$. There are at least $N^2/256$ of them for large N, each has $T_a \geq (4a-1)M_a \geq (N/2)M_a$, and all three sizes are at least $N/4-2$. Inserting the assumed lower bound for M_a, M_{b_1}, M_{b_2} gives $u_N \geq c'_\kappa N^{-(3\kappa-3)}$ for large N. Since $3/2-(3\kappa-3) = 3(3/2-\kappa)$, the distance to $3/2$ triples at each step, so after finitely many steps the exponent is below 1. That contradicts $u_n \leq \mu/(8n)$. □

So if a power law $u_n = n^{-\alpha+o(1)}$ holds, then $\alpha \geq 3/2$. Without that assumption, Theorem 9 says only that no lower bound $c_\kappa n^{-\kappa}$ with $\kappa < 3/2$ holds for all large n; it does not give $u_n = O(n^{-3/2})$. Heuristically, $3/2$ is the limit of the method: Theorem 8 is nearly an equality when u_n is of order $n^{-3/2}$, which is the behavior of tree-like families built only by insertion and products. Going further needs an input that separates meanders from such families. One candidate is that a positive fraction of meanders of order n has no capped block with more than n crossings; with it the same argument gives $\kappa < 2$. In the exact data this fraction is 0.5 at $n = 2$ and 3 and about 0.85 at $n = 24$ and 25, but it is not proved to stay positive.

4 The lower bound

The lower bound needs the opposite of (2): a large meander must not be much more common than a pair of smaller ones. The proof has four steps. A closed curve, the *separator*, cuts every meander into two pieces of comparable size and meets the road and the river only about $\sqrt{N}$ times (Section 4.1). Each piece is completed to a meander in a way that can be undone with little extra information (Section 4.2). This gives a reverse inequality $M_N \le e^{B\sqrt{N}} M_a M_b$ with a, b comparable to $N/2$ (Section 4.3). Repeated splitting, together with subadditivity, turns it into $L(n) \le C\sqrt{n}$ (Section 4.4).

The $\sqrt{n}$ has a plain source. The cost of the cut is exponential in the number of times the separator meets the road and the river. Some planar graphs really need about $\sqrt{N}$ such meetings to be cut into two pieces of comparable size; a square grid of side $\sqrt{N}$ is the standard example. Since the argument is applied to every meander, it must pay for the worst case. Whether a typical meander can be cut more cheaply is not known; Section 7 states what a cheaper cut would give.

4.1 A balanced separator

Close the river through a point at infinity, so that river and road become two closed curves on the sphere that cross at $2N$ points. Together they form a *planar map*, a graph drawn on the sphere, with $2N$ vertices, each of degree four. A *separator* is a closed curve on the sphere that avoids the $2N$ crossings and crosses the river and the road transversally. The points where it crosses them are called *ports*. It splits the sphere into two disks.

Proposition 10 (single balanced separator)**.** *For every sufficiently large N, every meander of order N has a separator with at most $4\sqrt{27N}$ ports such that each of its two disks contains between $N/2$ and $3N/2$ of the $2N$ crossings.*

The proof applies published theorems rather than proving a new one. Subdivide every edge of the map once, which gives a simple connected plane graph with $6N$ vertices, none of degree one. Its *branchwidth*, a standard measure of how cheaply a graph can be cut up, is at most $\sqrt{4.5 \cdot 6N} = \sqrt{27N}$ by a theorem of Fomin and Thilikos [10], which bounds the branchwidth of a planar graph with v vertices by $\sqrt{4.5v}$. By Seymour and Thomas [24] and Gu and Tamaki [12], in the form given by Dorn, Penninkx, Bodlaender and Fomin [7], such a graph has a *sphere-cut decomposition* of the same width. This is a tree whose leaves are the edges of the graph and in which every tree edge corresponds to a closed curve on the sphere that meets the graph in at most $\sqrt{27N}$ vertices and separates the graph edges on the two sides of the tree edge. Give each original crossing weight one, placed on a leaf at that crossing; a weighted centroid of the tree has an incident edge whose curve splits the weight between $2N/3$ and $4N/3$. Finally push the curve off each vertex it passes through; this creates at most four ports per vertex. Balanced separators of order $\sqrt{N}$ are classical for planar graphs [16, 19]; what matters here is that this one is a single closed curve, so that each side is one disk and all loose ends of a side lie on one circle.

4.2 Disk completion with a protective cycle

Cut the sphere along the separator. Each side is a *disk patch*: some v crossings, pieces of river and road inside, and w ports on the boundary circle. The river and the road each have a

positive even number of ports, since each closed curve visits both sides. The task is to turn a patch into a meander from which the patch can be read off again.

Proposition 11 (disk completion)**.** *There is an absolute constant c_Ω with the following property. Every disk patch with v crossings and w ports can be completed to a meander with at most $v + 8w$ crossings, in such a way that the patch, with the cyclic order of its ports, is recovered from the completed meander, a marked crossing of it, and a* record *taking at most $e^{c_\Omega w}$ possible values.*

Intuitively, the completed meander does not show where the patch sits inside it; the marked crossing and the record say where to find it. The construction has two parts, a cycle and caps, and the recovery uses the cycle; Figure 2 sketches it.

The protective cycle. Just outside the boundary, draw a closed curve Ω around the patch that crosses each port strand once. Call the river and the road the two *colors*; a crossing always joins strands of different colors. So along a stretch of Ω that crosses road strands, Ω is made of river, and along a stretch that crosses river strands, Ω is made of road. Where the color of the ports changes, Ω switches color: its river part and its road part cross at one new crossing, both continue outward past it, and Ω turns there from one to the other. So Ω is a cycle of the new map with at most $2w$ vertices: at a port it goes straight through, and at a switch it turns.

Caps. Outside Ω, every piece of river and every piece of road is an arc with two loose ends, because no closed curve can lie entirely in one patch. For each color, join the loose ends outside by a noncrossing family of arcs, the *caps*, nested at most two deep and chosen so that each color closes into a single curve. This can always be done by scanning the ends once around the circle and merging pieces one at a time. There are at most $3w$ loose ends: one for each port, and two for each switch. Each crossing between caps of different colors can be charged to an end of one color lying under a cap of the other color, and since caps are nested at most two deep, each end carries at most two charges. So the caps add at most $6w$ crossings. Together with the at most $2w$ vertices of Ω, the completion has at most $v + 8w$ crossings.

Recovery. Mark a vertex of Ω and record, step by step, how Ω continues: at each vertex of a map of degree four there are at most three ways to continue without going back, so the record has at most 3^{2w} possibilities. One more bit says which side of Ω holds the patch. The record recovers Ω. At a vertex where Ω goes straight, one strand enters the inside: that is a port. At a switch no strand enters. Deleting Ω and everything outside recovers the patch and the cyclic order of its ports.

The point of the protective cycle is that it is *recognizable*. Without it, an earlier completion had to record which edges the cut had crossed, which takes $N^{O(w)}$ possible records instead of $e^{c_\Omega w}$. With w of order $\sqrt{N}$, that is the difference between $e^{O(\sqrt{N}\log N)}$ and $e^{O(\sqrt{N})}$.

4.3 The reverse inequality

Proposition 12 (reverse inequality)**.** *There is a constant B such that for every sufficiently large N*

$$M_N \;\le\; e^{B\sqrt{N}} \max M_a M_b,$$

the maximum over integers a, b with $N \le a + b \le N + 8W$ and $N/10 \le a, b \le 9N/10$, where $W = \lceil 4\sqrt{27N} \rceil$.

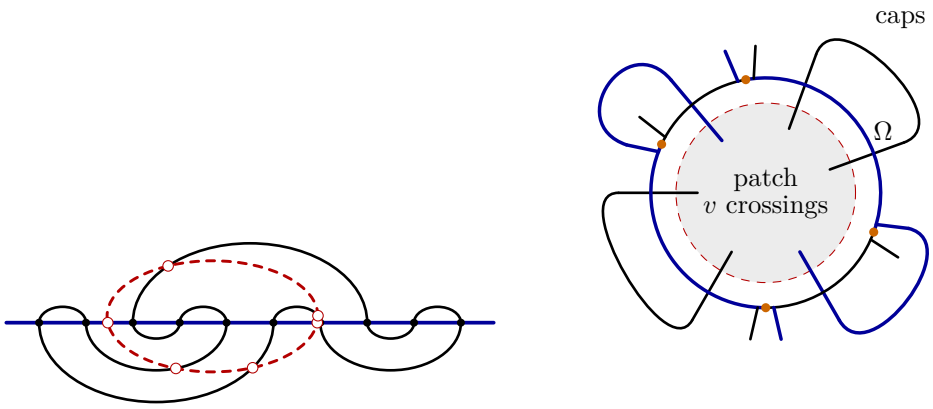


Figure 2: (a) The meander of Figure 3 with a separator (dashed). Its ports, the points where it crosses the river or the road, are circled. (b) Completion of one side, schematically: the patch inside the separator, the protective cycle Ω just outside it (river-colored where it crosses road strands, road-colored where it crosses river strands, switching color at the dots), and caps outside that close each color into one curve. The caps shown are only indicative.

Key idea. Choose, for each meander of order N, one separator as in Proposition 10, with $w \le W$ ports, and complete both sides as in Proposition 11. The sides have $v_1 + v_2 = 2N$ crossings with $N/2 \le v_i \le 3N/2$, and the completed meanders have orders a, b with $v_i/2 \le a, b \le v_i/2 + 4w$. The original meander is recovered from the two completed meanders with one marked crossing each, their two records, and the way the two boundary circles are glued, which is an orientation and a rotation, at most $2w$ choices. The marks, the sizes, and the choice of which crossing of the glued map is crossing 1 of the original meander, and in which direction the river runs, cost only a fixed polynomial in N. So the number of meanders of order N is at most a polynomial in N times $e^{2c_\Omega W}$ times $\max M_a M_b$, and the polynomial is absorbed into $e^{B\sqrt{N}}$. The ranges of a, b follow from those of v_i for large N. □

No ratio bound is used: the sizes a and b are allowed to exceed N in total by $O(\sqrt{N})$, and that excess is handled in the next step by $\mu \le 16$.

4.4 The descending chain

Proof of the lower bound in Theorem 1. By Proposition 12, for every large N there are $a, b \in [N/10, 9N/10]$ with

$$L(a) + L(b) \;\le\; L(N) + (a + b - N)\log\mu + B\sqrt{N} \;\le\; L(N) + B_1\sqrt{N},$$

using $a + b - N \le 8W$ and $\mu \le 16$. Divide by $a + b \ge N$ and use $L(N) \ge 0$: one of the two, call it k and call it a *child* of N, satisfies

$$\frac{L(k)}{k} \;\le\; \frac{L(N)}{N} + \frac{B_1}{\sqrt{N}}. \tag{6}$$

Intuitively, passing to a child increases the loss per crossing, $L(k)/k$, only slightly. Fix a large target n and start from a huge N_0. Pass repeatedly to a child, and stop at the first size $k \le n$;

then $n/10 < k \le n$. Each step shrinks the size by a factor between $1/10$ and $9/10$, so, read backward from the last one, the sizes grow at least geometrically and the added terms $B_1/\sqrt{\text{size}}$ sum to at most a constant times $1/\sqrt{n}$. Choosing N_0 so large that $L(N_0)/N_0 \le 1/\sqrt{n}$ gives a k in $(n/10, n]$ with $L(k) \le B_2\sqrt{n}$, for every large n.

Now take any n. Split off such a size $k_1 \in (n/10, n]$, then such a size k_2 for the remainder $n - k_1 \le 9n/10$, and so on. By subadditivity (3), $L(n)$ is at most $L(k_1) + L(k_2) + \cdots$ plus a bounded term for the last small remainder. Each $L(k_i)$ is at most B_2 times the square root of a geometrically decreasing size, so the sum is at most a constant times $\sqrt{n}$. This gives $L(n) \le C\sqrt{n}$ for large n, and enlarging C covers small n. □

The same descent turns a reverse inequality $M_N \le e^{O(N^\tau)} \max M_a M_b$, for a fixed $0 < \tau < 1$, into $L(n) = O(n^\tau)$, provided that the maximum runs over sizes a, b between two fixed fractions of N with $a + b - N = O(N^\tau)$. The excess $a + b - N$ costs $(a + b - N) \log \mu$ in the first display of the proof, and under these conditions the geometric sums still converge. Both conditions are needed: with the excess $8W$ of Proposition 12, which is of order $\sqrt{N}$, the descent gives only $L(n) = O(\sqrt{n})$, however small the factor $e^{O(N^\tau)}$ is. With the earlier completion the loss was of order $\sqrt{N} \log N$, which gave the previous bound. The argument also extends, with the same square root, to systems of several closed roads weighted by a parameter for each road; we do not use this here.

5 Products, primes and cuts

5.1 Cuts

Definition 13. A meander of order N has a *cut* at m, for $1 \le m \le N - 1$, if no lower arch passes over gap $2m$, the crossings $2m$ and $2m+1$ are joined by a short upper arch, and exactly one other upper arch passes over gap $2m$.

Intuitively, a cut is a thin neck: the road passes from the left part of the river to the right part exactly twice, once along the short arch and once along the arch over it. Figure 3 shows an example. The product of Section 2 creates exactly such a neck.

Theorem 14 (products and cuts)**.** *The product $(G, H) \mapsto G \cdot H$ of Section 2 is a bijection from pairs of meanders of orders m and $N - m$ onto meanders of order N with a cut at m. Moreover the cuts of $G \cdot H$ are the cuts of G, the position m, and the cuts of H shifted by m.*

Key idea. The product creates the short arch $\{2m, 2m+1\}$ and the arch $\{x, y\}$ over it, and no lower arch passes over the gap, so it has a cut at m. Conversely, at a cut, replacing the two upper arches over the gap, $\{x, y\}$ and $\{2m, 2m+1\}$, by $\{x, 2m\}$ and $\{2m+1, y\}$ separates the road into two loops, one on each side, and these are the factors. A gap $2m'$ with $m' < m$ sees the same lower arches as in G and the same upper arches except that $\{x, 2m\}$ became $\{x, y\}$, which passes over gap $2m'$ exactly when $\{x, 2m\}$ did; so cuts away from m are those of the factors. □

So exactly $M_m M_{N-m}$ meanders of order N have a cut at m, and for a uniformly random meander

$$\Pr_N(\text{cut at } m) \;=\; \frac{M_m M_{N-m}}{M_N} \;=\; \frac{u_m u_{N-m}}{u_N}. \tag{7}$$

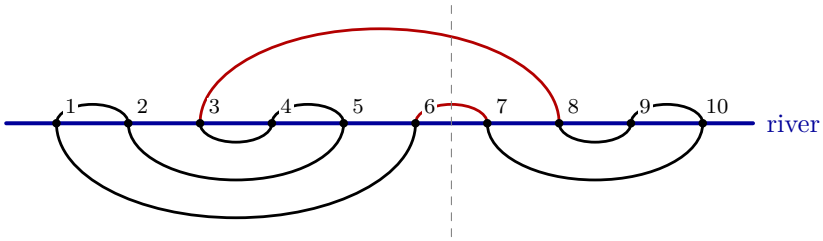


Figure 3: A meander of order 5 with a cut at $m = 3$ (dashed line, gap 6). No lower arch passes over the gap; above it there is the short arch 6–7 and one other arch, 3–8 (both red). Replacing these two arches by 3–6 and 7–8 splits the road into the running example of Figure 1 on crossings 1–6 and a meander of order 2 on crossings 7–10.

In particular $\Pr_{2m}(\text{cut at } m) = e^{-D(m)}$: the doubling defect of Section 2 is minus the logarithm of the probability of a cut at the middle.

5.2 Prime meanders

A meander is *prime* if it has no cut. By Theorem 14 and induction on the smallest cut, every meander is a product $Q_1 \cdot Q_2 \cdots Q_k$ of primes in exactly one way, and the product is associative. With P_n the number of prime meanders of order n, $\mathcal{M}(z) = \sum_{n\geq 1} M_n z^n$ and $\mathcal{P}(z) = \sum_{n\geq 1} P_n z^n$, this says

$$1 + \mathcal{M}(z) \;=\; \frac{1}{1 - \mathcal{P}(z)}.$$

The first values are $P_n = 1, 1, 5, 27, 175, 1255, 9699, 79253, 676429, \ldots$, and $P_{32}/M_{32} = 0.794$. Write $U = \sum_n u_n = \mathcal{M}(1/\mu)$, which is finite by Theorem 15 below. Then $\mathcal{P}(1/\mu) = U/(1 + U) \leq 1/5$. In the language of renewal theory, (u_n) is a *transient* renewal sequence: a meander is a sequence of prime blocks, and the weights $P_n \mu^{-n}$ of the possible blocks add up to less than one. That a positive fraction of meanders is prime is not proved.

5.3 The sum of u_n

Let X_N be the number of meanders of order N with one marked exterior upper arch (one not enclosed by another upper arch), so that $M_N \leq X_N \leq N M_N$. By reflection in the river, X_N also counts meanders with a marked exterior lower arch.

Theorem 15. $\sum_{n\geq 1} u_n \leq 1/4$.

Proof. Two facts are needed. First, the gaps other than the two ends over which no lower arch passes are exactly the gaps between consecutive exterior lower arches, so a meander with η exterior lower arches has $\eta - 1$ of them. Second, the meanders of order N in which no lower arch and exactly two upper arches pass over gap $2m$ number exactly $X_m X_{N-m}$. Indeed, if the two upper arches are $\{p, r\}$ and $\{p', r'\}$ with $p < p' \leq 2m < r' < r$, replacing them by $\{p, p'\}$ and $\{r', r\}$ gives a meander on each side, each with a marked exterior arch, and this is reversible; the case $p' = 2m$, $r' = 2m + 1$ is a cut.

Counting only the gaps over which no lower arch and exactly two upper arches pass, we get

$$X_N - M_N \;=\; \sum_{\Gamma}(\eta(\Gamma)-1) \;\geq\; \sum_{m=1}^{N-1} X_m X_{N-m},$$

where the sum is over meanders Γ of order N and $\eta(\Gamma)$ is the number of exterior lower arches. With $\mathcal{X}(z)=\sum_n X_n z^n$ this says that $\mathcal{X}-\mathcal{X}^2-\mathcal{M}$ has nonnegative coefficients. For $0 \leq z < 1/\mu$ all three series converge, and $\mathcal{M}(z) \leq \mathcal{X}(z) - \mathcal{X}(z)^2 \leq 1/4$. Letting z increase to $1/\mu$ gives $\sum_n M_n \mu^{-n} \leq 1/4$ by monotone convergence. □

In particular $u_n \to 0$, and for every $\xi \in (0,1]$ at most $N^{\xi}/4$ of the $n \leq N$ have $u_n > n^{-\xi}$. From the exact counts and $\mu \leq 12.2970$, $U \geq \sum_{n\leq 32} M_n\, 12.297^{-n} > 0.1032$, so

$$0.1032 \;\leq\; U \;\leq\; 0.25.$$

A numerical estimate, which assumes the power law (1) for the tail, is $U \approx 0.1037$.

5.4 The number of cuts

By (7) the expected number of cuts of a random meander of order N is

$$\mathbb{E}_N[\#\text{cuts}] \;=\; \sum_{m=1}^{N-1} \frac{M_m M_{N-m}}{M_N}.$$

Every cut is a gap over which no lower arch passes, so the number of cuts is at most $\eta - 1$, where η is the number of exterior lower arches. The mean number of exterior upper arches, which equals the mean number of exterior lower arches, is X_N/M_N, and a short surgery at the marked exterior arch shows $2X_N \leq M_{N+1}$ [18]. Hence

$$\mathbb{E}_N[\#\text{cuts}] \;\leq\; \frac{M_{N+1}}{2M_N} - 1 \;\leq\; \frac{R_\star}{2} - 1, \tag{8}$$

which is bounded by Theorem 2. The exact values are 0.338, 0.270 and 0.245 at $N = 10$, 20 and 32, decreasing (Table 2).

5.5 Cuts away from the ends

Proposition 16. *There is a constant C_{c} such that for all $N \geq 2$ and $1 \leq k \leq N-1$, with $s = \min(k, N-k)$,*

$$\Pr_N(\textit{cut at } k) \;\leq\; C_{\mathrm{c}}\, s^{-1/4}\sqrt{\log(s+1)}.$$

Key idea. Write $\psi(k) = M_k M_{N-k}/M_N$, the probability of a cut at k; it is symmetric, so let $k \leq N/2$. Moving the cut by t positions changes $\log\psi$ by a difference of two sums of t terms $\log(M_{i+1}/(\mu M_i))$, since the factors μ cancel. By Theorem 3 each term is at most $\delta(i) = 2C_{\mathrm{r}}\, i^{-1/4}\sqrt{\log(i+1)}$ in absolute value, with all indices at least $k/2$. So for $t_0 = \lfloor 1/\delta(\lceil k/2\rceil)\rfloor$, about $k^{1/4}$, there are t_0+1 positions where the probability of a cut is at least $e^{-2}\psi(k)$. Their sum is at most $\mathbb{E}_N[\#\text{cuts}]$, which is bounded by (8), so $\psi(k) = O(\delta(\lceil k/2\rceil))$. Bounded k are handled by enlarging the constant. □

Intuitively, the probability of a cut is nearly constant over stretches of about $s^{1/4}$ positions, and the total over all positions is bounded. So no neck far from both ends carries a positive fraction of meanders. Under the power law the true order is $(N/(k(N-k)))^{\alpha}$, much smaller. The bound is not summable over positions, and a better ratio rate would not make it summable, so it does not settle whether the cuts in the middle carry vanishing total probability (Section 7.4).

5.6 Near-cuts

A cut is a very special neck. A meander of order $2m$ can also be nearly split at the middle in looser ways. These looser events are the ones that remain visible when a large random meander is rescaled to a continuum picture, as in Paper D [21]. Two results compare them with exact cuts.

Zero defect. Say that a meander of order $2m$ is *split at the middle* if its first $2m$ crossings form one stretch of consecutive crossings along the road. On a closed road, a set of crossings is one stretch exactly when two road arches leave it, so this is the event that exactly two arches pass over gap $2m$. Both arches are upper or both are lower, by parity, and the count in the proof of Theorem 15 gives X_m^2 for each case.

Proposition 17 (zero defect)**.** $\Pr_{2m}(\textit{split at the middle}) = 2X_m^2/M_{2m}$ *exactly. Consequently, since* $M_m^2/M_{2m} = e^{-D(m)}$,

$$2e^{-D(m)} \;\le\; \Pr_{2m}(\textit{split at the middle}) \;\le\; \tfrac{1}{2}R_\star^2\, e^{-D(m)}.$$

The inequalities follow from $M_m \le X_m \le M_{m+1}/2 \le R_\star M_m/2$. So a split at the middle is a cut at the middle up to a bounded factor: given a split, the probability of an exact cut is at least $2/R_\star^2$.

Bounded defect. More generally, say the first $2m$ crossings have *defect at most* K if they differ in at most K crossings from some stretch of $2m$ consecutive crossings along the road. Such a meander has a *near-cut* at the middle. Each changed crossing adds at most two road arches leaving the set, so at most $2+2K$ arches pass over gap $2m$. The next result bounds such meanders by exact products.

Proposition 18 (linear-record completion)**.** *For all* $m, n \ge 1$ *and even* $w \ge 2$*, the number of meanders of order* $m+n$ *in which at most* w *arches pass over gap* $2m$ *is at most* $(64R_\star^3)^w M_m M_n$.

Key idea. Suppose exactly w arches pass over the gap, $w_\uparrow$ upper and $w_\downarrow$ lower; smaller numbers are handled in the same way. Cut at the gap. The left half has $2m$ crossings and w loose ends. Append w new crossings on the right and join the loose ends to them in nested order; this closes the half into a system of at most w closed roads, each of which meets the new crossings. Then merge the roads from left to right along the new crossings: at each of the $w-1$ steps, either the next new crossing is already on the current road, or one of three local moves joins the two roads, adding at most two crossings. The record of the whole merge is a word of length $w-1$ in four letters, one letter per step. Because the steps follow a fixed left-to-right schedule, the word determines where each move happened, so the half is recovered from the completed meander and the word. This gives at most $4^{w-1}R_\star^{3w/2}M_m$ left halves with given $w_\uparrow$ and $w_\downarrow$, the factor $R_\star^{3w/2}$ paying for the at most $3w/2$ extra orders. The same holds on the right, and the two halves are glued across the gap in only one way. □

Corollary 19. *For every $K \geq 0$,*

$$e^{-D(m)} \;\leq\; \Pr_{2m}(\textit{defect at most } K) \;\leq\; (64R_\star^3)^{2K+2}\, e^{-D(m)}.$$

So near-cuts with defect K are at most a factor $e^{O(K+1)}$ more likely than exact cuts. For a slowly growing defect, $K = o(\log m)$, the two probabilities agree up to a factor $m^{o(1)}$, so they decay with the same power of m if either decays like a power.

6 Meanders with bounded labels

For all meanders the exponent is out of reach. For one natural subfamily both the growth rate and the exponent can be found exactly, and the answer shows a mechanism that is believed to produce a power law in general: a walk confined to a wedge.

6.1 Labels

In this section m is the order of a meander. Give each crossing $i < 2m$ an integer label φ_i so that along every arch above the river the label drops by one from left to right, and along every arch below the river it rises by one. Precisely: $\varphi_1 = 0$, and $\varphi_y = \varphi_x - 1$ for every upper arch $\{x, y\}$ with $x < y < 2m$, and $\varphi_y = \varphi_x + 1$ for every lower arch $\{x, y\}$ with $x < y < 2m$; the two arches at the last crossing $2m$ are left out. Removing them leaves the road as a single path through crossings $1, \ldots, 2m-1$, so these rules determine the labels uniquely. Intuitively, φ_i counts the half turns of the road on its way from crossing 1 to crossing i, with opposite signs above and below the river. For a meander, labels of neighboring crossings along the river differ by one. Conversely, a pair of arch systems is a meander if and only if it has a labeling that obeys these rules and in which neighboring labels differ by one. Paper B explains these labels [22]. In the running example of Figure 1 the labels of crossings 1 to 5 are $0, -1, 0, 1, 0$.

Let $M_m^{[-1,2]}$ be the number of meanders of order m whose labels all lie in $\{-1, 0, 1, 2\}$. It starts $1, 2, 7, 31, 161, 932, 5841, 38891$. The labels depend on which crossing is first and which is last, so this is a family of meanders with distinguished ends, not closed under all symmetries of the picture.

6.2 The growth rate

Let $x_0 = 0.9175433408\ldots$ be the positive root of $4x^4 = 2x + 1$ and

$$\Lambda \;=\; (1 + 2x_0)(1 + x_0)^2 \;=\; 10.4245356612\ldots,$$

the larger real root of the irreducible polynomial $16x^4 - 112x^3 - 520x^2 - 536x + 25$.

Theorem 20 (growth rate)**.** *There is a constant C_Λ such that for every $m \geq 1$*

$$\Lambda^{m-1} e^{-C_\Lambda \sqrt{m}} \;\leq\; M_m^{[-1,2]} \;\leq\; \Lambda^{m-1}.$$

Key idea. Read the crossings from left to right. After crossing j the state is the current label together with the labels of the upper arches and of the lower arches that are still open, that is, started but not yet closed. The open arches on each side are closed in last-in-first-out order, so they form a stack. Since the labels take only four values, each stack is a word in a finite

alphabet, and the reading process is a machine with two stacks and finitely many control states. This machine has an explicit eigenfunction of product form, with one weight for each stack entry below the top, and normalizing by it turns the count into a probability: $M_m^{[-1,2]} = \Lambda^{m-1}$ times the probability that an explicit Markov chain, reading one crossing at a time, is back in its starting state after $2m-2$ crossings without having been killed (stopped) on the way. The upper bound follows at once. For the lower bound, consider the chain without the constraint that a stack cannot go below empty. It has an invariant law of product form, and under it two explicit martingales show that with probability at least $1/2$ the chain stays for t steps within windows of length about $\sqrt{t}$ at the tops of the two stacks. Building such windows from the start and removing them at the end costs a factor $e^{-O(\sqrt{t})}$. □

6.3 The exponent

Let $\theta = \arccos\big((1+2x_0)/(3+4x_0)\big) = 1.13179\ldots$, which is $64.8468\ldots$ degrees, and

$$\alpha_{\mathrm{b}} \;=\; 1 + \frac{\pi}{\theta} \;=\; 3.7757713065\ldots.$$

Theorem 21 (exponent)**.** *For every $\epsilon > 0$ there are $c_\epsilon, C_\epsilon > 0$ such that for every $m \geq 2$*

$$c_\epsilon\, \Lambda^m m^{-\alpha_{\mathrm{b}}-\epsilon} \;\leq\; M_m^{[-1,2]} \;\leq\; C_\epsilon\, \Lambda^m m^{-\alpha_{\mathrm{b}}+\epsilon}.$$

That is, $M_m^{[-1,2]} = \Lambda^m m^{-\alpha_{\mathrm{b}}+o(1)}$.

Both numbers are explicit, and neither is rational: Λ is algebraic of degree four, and α_{b} is irrational, indeed transcendental. For the irrationality, $u = 2\cos\theta$ is a root of the polynomial

$$113u^4 - 312u^3 + 312u^2 - 128u + 16,$$

obtained by substituting $x_0 = (2-3u)/(4u-4)$ into $4x_0^4 - 2x_0 - 1 = 0$. It is irreducible: its reduction modulo 3, made monic, is $X^4 - X - 1$, which has no factor of degree one or two over the field with three elements. So the monic minimal polynomial of u has the constant term $16/113$, and u is not an *algebraic integer*, a root of a monic polynomial with integer coefficients. If θ/π were rational, $e^{i\theta}$ would be a root of unity and $u = e^{i\theta} + e^{-i\theta}$ would be an algebraic integer. Hence θ/π is irrational, and so is $\alpha_{\mathrm{b}} = 1 + \pi/\theta$. Since $e^{i\theta}$ is algebraic, the Gelfond–Schneider theorem then makes θ/π, and with it α_{b}, transcendental. The sharp form $M_m^{[-1,2]} \asymp \Lambda^m m^{-\alpha_{\mathrm{b}}}$, with bounded ratio, is not proved, and no closed formula for the counts is known.

6.4 The generating function

A power series is *D-finite* if it satisfies a nonzero linear differential equation whose coefficients are polynomials. Algebraic power series are D-finite, but many D-finite series are not algebraic. The generating function of the bounded-label family is in neither class.

Corollary 22. *The generating function*

$$F_{\mathrm{b}}(z) \;=\; \sum_{m \geq 1} M_m^{[-1,2]} z^m$$

is not D-finite, even with complex polynomial coefficients. In particular, it is not algebraic.

Proof. By the previous subsection, α_{b} is irrational. Choose an integer k with $k+1>\alpha_{\mathrm{b}}$. The coefficient bounds of Theorem 21 give

$$\lim_{r\downarrow 0}\frac{\log F_{\mathrm{b}}^{(k)}\big((1-r)/\Lambda\big)}{\log(1/r)} \;=\; k+1-\alpha_{\mathrm{b}}. \tag{9}$$

To see this, note that the kth derivative of z^m is $m(m-1)\cdots(m-k+1)\,z^{m-k}$, and this factor is of order m^k for large m. So for each small $\epsilon>0$ the positive series $F_{\mathrm{b}}^{(k)}((1-r)/\Lambda)$ lies, up to bounded factors and finitely many bounded terms, between $\sum_m m^{k-\alpha_{\mathrm{b}}-\epsilon}(1-r)^m$ and $\sum_m m^{k-\alpha_{\mathrm{b}}+\epsilon}(1-r)^m$. These sums are of order $r^{-(k+1-\alpha_{\mathrm{b}}\mp\epsilon)}$, which tends to infinity. Taking logarithms and letting ϵ decrease to zero proves (9).

Suppose now that F_{b} were D-finite. Its coefficients are rational, so an equation with complex coefficients can be replaced by one with rational coefficients: for a fixed order and degree, the coefficients of the equation solve a system of homogeneous linear equations with rational coefficients, and such a system with a nonzero complex solution has a nonzero rational one. Since the coefficients of F_{b} are integers bounded by 16^m, it is then a *G-function*, a D-finite series whose coefficients satisfy standard arithmetic growth bounds. The André–Chudnovsky–Katz theorem says that the minimal equation of a G-function has only regular singularities, with rational exponents [9, Theorem 6]. By (9), $1/\Lambda$ is a singular point. So near $1/\Lambda$ the function and its derivatives are finite sums of rational powers of $1/\Lambda-z$ times powers of $\log(1/\Lambda-z)$ times analytic functions. After equal powers are combined, the first nonzero term shows that the limit in (9) is rational. This contradicts the irrationality of α_{b}. Finally, every algebraic power series is D-finite. □

The proof uses the coefficient bounds of Theorem 21 with arbitrary ϵ; it does not need the sharp form $M_m^{[-1,2]}\asymp\Lambda^m m^{-\alpha_{\mathrm{b}}}$. It also needs nothing about other singularities on the circle $|z|=1/\Lambda$, because all estimates are made on the positive real axis.

6.5 The wedge

The following heuristic explains why the exponent is $1+\pi/\theta$; the proofs make it rigorous. Let $h_{\uparrow}(j)$ and $h_{\downarrow}(j)$ be the numbers of upper and lower arches still open after crossing j: the heights of the two stacks. A meander starts and ends with both heights zero, and in between neither is negative. So the pair of heights is a walk in the quarter plane that starts and ends at the corner.

For the chain of Theorem 20, the two heights after t steps, rescaled by $\sqrt{t}$, converge to a planar Brownian motion whose two coordinates have correlation

$$\rho \;=\; -\frac{1+2x_0}{3+4x_0} \;=\; -0.42504\ldots.$$

A linear change of coordinates turns this Brownian motion into a standard one and the quarter plane into a wedge of angle $\theta=\arccos(-\rho)$. The correlation is negative: when one stack grows, the other tends to shrink, so it is harder for both to stay positive, and the wedge is narrower than a right angle.

For standard Brownian motion in a wedge of angle θ, started at distance one from the corner, the probability of staying inside for time t decays like $t^{-\pi/(2\theta)}$ [5]. A meander asks for more: an excursion from the corner back to the corner. That costs $t^{-\pi/(2\theta)}$ for the first

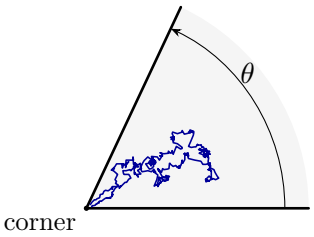


Figure 4: Schematic: after a linear change of coordinates the two stack heights move like a planar Brownian motion in a wedge of angle θ, and a meander is an excursion from the corner back to the corner. Here $\theta \approx 64.85°$. The path drawn is illustrative, not a sample.

half, the same for the second half read backward, and t^{-1} for ending in a bounded region near the corner, as for any two-dimensional walk. Altogether $t^{-1-\pi/\theta} = t^{-\alpha_{\mathrm{b}}}$, and $t = 2m - 2$ is proportional to the order. Figure 4 shows the picture.

Two checks make the formula plausible. Without the requirement of a single loop, the upper and lower arch systems are independent, the heights are independent, $\rho = 0$, $\theta = \pi/2$, and the exponent is $1+2 = 3$; indeed the number of pairs of arch systems is $\mathrm{Cat}_n^2 \approx 16^n/(\pi n^3)$. For the narrower band of labels $\{-1, 0, 1\}$, the counts $1, 2, 6, 22, 92, 422, 2074, \ldots$ are the Baxter numbers. Chung, Graham, Hoggatt and Kleiman gave an explicit formula for them [4], from which they grow like a constant times $8^n n^{-4}$; there $\rho = -1/2$, $\theta = \pi/3$ and $1 + \pi/\theta = 4$. The identification with the Baxter numbers is from the literature and was checked here only on small orders, not proved.

The main difficulty in the proofs is that the heights are not a random walk with independent steps: the chain is Markov only on its full state, two words of unbounded length. The proofs replace independence by stationarity under the invariant law, and the lower bound also uses that the chain reversed in time is again an explicit two-stack machine.

For all meanders the stacks are unbounded in both directions and no finite description of this kind is known. Whether the exponent of all meanders arises from a wedge in a similar way is the question behind Papers A and D [23, 21].

7 What would improve the bounds

Theorem 1 leaves the loss $L(n)$ between about $\log n$ and $C\sqrt{n}$. This section states what would lower the upper estimate $C\sqrt{n}$, why some natural routes cannot, and what weaker targets remain. The propositions and the argument that gluing cannot help have complete written proofs, which have not been checked independently; the expected value of β is a heuristic.

7.1 Short separators for a fraction of meanders

The proof of the lower bound applies one separator with about $\sqrt{N}$ ports to every meander. A separator with fewer ports may exist only for some meanders; then the argument must also pay for how rare these meanders are.

Proposition 23 (fraction criterion)**.** *Let $0 < \tau < 1$ and $c_{\mathrm{f}} > 0$. Suppose that for all large N a fraction at least $e^{-c_{\mathrm{f}} N^{\tau}}$ of the meanders of order N have a separator with at most $c_{\mathrm{f}} N^{\tau}$ ports*

whose two disks each contain between $N/2$ and $3N/2$ of the $2N$ crossings. Then $L(n) \le K_{\mathrm{f}}\, n^{\tau}$ for all n, for some constant K_{f}.

Key idea. Run the proof of Proposition 12 on these meanders only. Their number is at least $e^{-c_{\mathrm{f}} N^{\tau}} M_N$ and at most a polynomial times $e^{2c_\Omega c_{\mathrm{f}} N^{\tau}} \max M_a M_b$, so $L(a)+L(b) \le L(N)+B'N^{\tau}$. The descent of Section 4.4 goes through with $N^{\tau-1}$ in place of $N^{-1/2}$, because the geometric sums still converge. □

With $\tau = 1/2$ and fraction one this is the lower bound of Theorem 1. The criterion allows separators of any shape. For example, if every face of the map can be reached from one fixed face by crossing at most ω edges, the map has a balanced separator with at most $2\omega+1$ ports.

7.2 Shallow meanders

The simplest balanced cut is a vertical line through the middle gap of the river. It always exists, it splits the crossings evenly, and its ports are the arches that pass over the gap, so its cost depends on how many arches do. Say that an arch system has *depth* at most d if no point lies under more than d of its arches, and let $M_n^{(d)}$ count meanders of order n whose upper and lower arch systems both have depth at most d. We call these *shallow* meanders.

Proposition 24 (depth criterion)**.** *Suppose that for some K_{d} and $\beta > 0$*

$$M_n^{(d)} \;\ge\; \mu^n \exp\bigl(-K_{\mathrm{d}}\, n\, d^{-\beta} - K_{\mathrm{d}}\, d\bigr) \qquad \text{for all } n, d \ge 1.$$

Then $L(m) \le K'_{\mathrm{d}}\, m^{1/(1+\beta)}$ for all m, for some constant K'_{d}.

Proof. In a meander of depth at most d, at most $2d$ arches pass over the middle gap. By Proposition 18, $M_{2m}^{(d)} \le e^{2B_{\mathrm{lin}} d} M_m^2$ with $B_{\mathrm{lin}} = \log(64R_\star^3)$. With the hypothesis at $n = 2m$,

$$2m \log\mu - 2K_{\mathrm{d}} m d^{-\beta} - K_{\mathrm{d}} d \;\le\; 2B_{\mathrm{lin}} d + 2\log M_m,$$

so $L(m) \le K_{\mathrm{d}}\, m\, d^{-\beta} + (B_{\mathrm{lin}} + K_{\mathrm{d}}/2)\, d$. Take $d = \lceil m^{1/(1+\beta)} \rceil$. □

The exponent $1/(1+\beta)$ is below $1/2$ exactly when $\beta > 1$. So shallow meanders improve on the square root if and only if confining a meander to depth d costs less than d^{-1} per crossing in the exponent.

Gluing short meanders never helps. Let $\mu_d = \lim_n (M_n^{(d)})^{1/n}$ be the growth rate of shallow meanders. The only unconditional lower bounds for $M_n^{(d)}$ come from building shallow meanders out of short ones, by products. A meander of order $n_1 \le d$ has depth at most n_1, and a product of many of them, combined in a balanced way, adds only a few levels of nesting. This gives $\log(\mu/\mu_d) \lesssim L(n_1)/n_1$ with $n_1 \approx d$. If $L(n_1) = O(n_1^{\tau})$, that is $\beta = 1-\tau$, and Proposition 24 returns the exponent $1/(2-\tau)$. Since

$$\frac{1}{2-\tau} - \tau \;=\; \frac{(1-\tau)^2}{2-\tau} \;\ge\; 0,$$

the exponent never improves; from $\tau = 1/2$ it gives $2/3$, worse than the square root. A proof along these lines must produce long, thin meanders that are not concatenations of short ones.

What is expected. Heuristically, $\beta \approx 1.92$. Typical arch heights in large meanders, the numbers of arches nested above a point, are predicted by a KPZ-type argument to grow like

n^ν with $\nu \approx 0.52$, and Monte Carlo measurements on the closely related semi-meanders, where the river is a half-line, give 0.518 [11]. For a process whose height after n steps is about n^ν, staying below height d for a long time costs a factor of about $\exp(-\text{const} \cdot d^{-1/\nu})$ per step, which suggests $\log(\mu/\mu_d) \approx \text{const} \cdot d^{-1/\nu}$ and $\beta = 1/\nu \approx 1.92$. If so, and if the hypothesis of Proposition 24 holds with this β, it gives $L(n) = O(n^{0.34})$. For comparison, pairs of arch systems without the single-loop condition have $\beta = 2$ exactly: those of depth at most d grow like $\sigma_d = (2\cos(\pi/(d+2)))^4$ per order, and $16 - \sigma_d \approx 32\pi^2/(d+2)^2$.

A cleaner sufficient condition. The hypothesis of Proposition 24 would follow, with β up to 2, from a correlation statement: that $\mu_d/\sigma_d \geq (\mu/16)\exp(-O(d^{-\beta'}))$ for some $\beta' > 1$, with uniform prefactors. In words, among pairs of arch systems of depth at most d, the probability of forming a single loop decays at nearly the same exponential rate as for all pairs. The growth rates μ_d can be computed exactly for small d by transfer matrices, finite matrices that build a shallow meander one crossing at a time, so this can be tested before a proof is attempted.

7.3 Existence of an exponent is not easier

One might hope to prove that some exponent exists, $u_n = n^{-\alpha+o(1)}$ for some finite α, without finding it. This is at least as hard as a lower bound on M_n of the form μ^n times a power of n: if such an α exists, then $L(n) \leq (\alpha+1)\log n$ for large n, that is $M_n \geq \mu^n n^{-\alpha-1}$. By Corollary 5, such a polynomial lower bound is the same as $D(m) \leq \chi \log m + O(1)$ for some χ, that is, $M_{2m} = O(m^\chi M_m^2)$. No method in this paper gives that.

7.4 A weaker target: subexponentiality

A summable positive sequence (a_n) is called *subexponential* if

$$\frac{1}{a_N}\sum_{m=1}^{N-1} a_m a_{N-m} \;\longrightarrow\; 2\sum_n a_n \qquad (N \to \infty).$$

In words, a large total is almost always made of one large part and one small part. Take $a_n = u_n$. By (7) the left side is exactly $\mathbb{E}_N[\#\text{cuts}]$. So

$$(u_n) \text{ is subexponential} \quad\Longleftrightarrow\quad \mathbb{E}_N[\#\text{cuts}] \to 2U.$$

Since $\Pr_N(\text{cut at } m) \to u_m$ for each fixed m, by Theorem 2, the cuts near the two ends already contribute $2U$ in the limit. So subexponentiality says that cuts with both sides large carry vanishing total probability: for every $\epsilon > 0$ there is k_0 with $\sum_{k_0 \leq k \leq N-k_0} \Pr_N(\text{cut at } k) \leq \epsilon$ for all large N. Intuitively, when a large meander has a cut, one side is almost always small.

This is a different target from a polynomial lower bound, and neither is known to imply the other. It allows fast decay: sequences like $e^{-\sqrt{n}}$ are subexponential. It would have a concrete consequence. Since every meander is a sequence of primes and $\sum u_n$ is finite, the standard theory of subexponential sequences [3] would give a positive limiting fraction of prime meanders, $P_n/M_n \to (1 - \mathcal{P}(1/\mu))^2$; the exact hypotheses of that theory remain to be checked here. The exact data are consistent with it: $\mathbb{E}_N[\#\text{cuts}]$ decreases, to 0.245 at $N = 32$, while $2U \geq 0.2064$ and the estimate of U gives $2U \approx 0.207$; the middle part with $k_0 = 5$ is 0.0127 at $N = 32$ (Table 2). Proposition 16 bounds each middle term, but its bound is not summable. What is missing is an argument that sees many cut positions at once; local smoothness of u_n alone cannot give it.

8 Open problems

1. *A polynomial lower bound.* Prove $M_n \geq \mu^n n^{-\chi}$ for some χ, equivalently $M_{2m} = O(m^{\chi'} M_m^2)$ for some χ'. This is the main gap. By Proposition 18, it would follow from a lower bound, polynomial in m, for the probability that at most a fixed number of arches pass over the middle gap of a meander of order $2m$. It is implied by the existence of an exponent.

2. *Improving the square root.* Prove $L(n) = o(\sqrt{n})$. Two routes are stated precisely: short balanced separators for a not too small fraction of meanders (Proposition 23), or the depth hypothesis of Proposition 24 with some $\beta > 1$. For the second, the correlation statement for shallow arch systems in Section 7 is a cleaner sufficient condition that can first be tested exactly.

3. *An upper bound beyond* $1/n$. Prove $u_n = O(n^{-\kappa})$ for some $\kappa > 1$. Theorem 9 gives $3/2$ only along a subsequence. A related open statement is $M_{n+1} \leq \mu M_n$ for every n, which holds for $n \leq 31$.

4. *Middle cuts.* Show that cuts with both sides large carry vanishing total probability, that is, that (u_n) is subexponential, and deduce that a positive fraction of meanders is prime.

5. *Bounded labels, sharp form.* Prove $M_m^{[-1,2]} \asymp \Lambda^m m^{-\alpha_{\mathfrak{b}}}$, and find the exponents of wider bands of labels. For every band of labels that contains 0 and 1, the corresponding family is closed under products, so it has a growth rate, but the exponents of the wider bands are not known, nor whether they approach α_* as the band widens.

6. *The exponent.* Prove that $u_n = n^{-\alpha+o(1)}$ for some α, and that $\alpha = \alpha_*$. Paper A explains the conjectured value [23], and Paper D studies a continuum counterpart [21].

A Exact data

All values below are computed from the published exact counts M_n for $n \le 32$: OEIS A005315 [20] for $n \le 28$ and [14] for $29 \le n \le 32$. They need no value of μ. Nothing is fitted.

m	$D(m)$	$m\,(D(m+1)-D(m))$	$D(2m)-D(m)$	m	$D(m)$	$m\,(D(m+1)-D(m))$
1	0.6931	1.6582	1.6582	9	6.5482	2.9622
2	2.3514	2.0014	1.7902	10	6.8773	3.0023
3	3.3521	2.3683	1.9612	11	7.1776	3.0361
4	4.1415	2.5426	2.0425	12	7.4536	3.0648
5	4.7772	2.6804	2.1001	13	7.7090	3.0896
6	5.3133	2.7781	2.1403	14	7.9466	3.1112
7	5.7763	2.8539	2.1703	15	8.1689	3.1302
8	6.1840	2.9137	2.1935			

Table 1: The doubling defect $D(m) = \log(M_{2m}/M_m^2) = -\log \Pr_{2m}(\text{cut at } m)$. If $u_n \sim An^{-\alpha}$, the column $D(2m) - D(m)$ tends to $\alpha \log 2$, and, if the correction is also smooth, the column $m\,(D(m+1) - D(m))$ tends to α; for α_* these limits are 3.4201 and 2.3707. The first needs $D(m+1)$, hence M_{2m+2}, and the second needs M_{4m}, so it stops at $m = 8$.

N	$\mathbb{E}_N[\#\text{cuts}]$	$k_0 = 2$	$k_0 = 3$	$k_0 = 5$	$10^4\, M_{N/2}^2/M_N$
10	0.3378	0.1088	0.0544	0.0084	84.20
12	0.3159	0.0994	0.0514	0.0156	49.26
16	0.2876	0.0858	0.0445	0.0178	20.62
20	0.2704	0.0770	0.0393	0.0166	10.31
24	0.2590	0.0710	0.0355	0.0151	5.79
28	0.2510	0.0668	0.0327	0.0138	3.54
32	0.2451	0.0636	0.0306	0.0127	2.30

Table 2: The expected number of cuts $\mathbb{E}_N[\#\text{cuts}] = \sum_{k=1}^{N-1} M_k M_{N-k}/M_N$, the part of this sum with $k_0 \le k \le N - k_0$, and the probability of a cut exactly at the middle (times 10^4). The total does not increase for any $N \le 32$; the middle parts with $k_0 = 2$, 3 and 5 decrease for $N \ge 6$, 9 and 15 respectively. If (u_n) is subexponential (Section 7.4), the total tends to $2U \ge 0.2064$.

The prime counts P_n follow from $1 + \mathcal{M} = 1/(1 - \mathcal{P})$, that is $M_n = \sum_{k=1}^{n} P_k M_{n-k}$ with $M_0 = 1$; $P_{32}/M_{32} = 0.7938$. The ratios M_{n+1}/M_n increase for $n \le 31$, from 2 to 11.0227.

B Where the proofs are

All proofs, programs and data are in the project repository [18], folder `meandric/research`. "Checked" means that a separate, independent reading of the written proof found no gap; the folder of that reading is given in parentheses. "Proved" means that a complete proof is written but has not been checked in this way. "Data" means exact computation, with no claim of proof. Nothing has been reviewed outside the project.

result	folders	status
Products and cuts, primes (Theorem 14)	120; also 403	checked (120, 404)
Ratio limit (Theorem 2)	113	checked (113)
Rate in the ratio limit (Theorem 3)	628, 621	checked (631)
Rate $n^{-1/4}$ (remark after Theorem 3)	1342, 621	checked (1340)
Halving identity (Proposition 4, Corollary 5)	621, 628; full proof here	proved
Insertion, $M_n \le \mu^{n+1}/(8n)$ (Theorem 6, Lemma 7)	403; constant 8 from 404	checked (404)
Three-piece insertion, exponent 3/2 (Theorems 8, 9)	403	checked (404)
$\sum u_n \le 1/4$ (Theorem 15)	120	checked (120)
Mean number of cuts (8)	120, 113	checked (120, 113)
Cuts away from the ends (Proposition 16); form with $s^{-1/4}$	review_2026-10-25_ideas_2_3; this paper	proved
Balanced separator (Proposition 10)	1340	checked (1342)
Disk completion (Proposition 11)	1340, caps in 1343	checked (1341, 1342)
Reverse inequality, $M_n \ge \mu^n e^{-C\sqrt{n}}$ (Proposition 12, Theorem 1)	1340	checked (1341)
Earlier bound with $\sqrt{n}\log n$	628	checked (631)
Extension to loop-weighted systems	1340	checked (1342)
Zero defect (Proposition 17)	1340	checked (1343)
Linear-record completion (Proposition 18, Corollary 19)	1343; earlier form 1340	checked (1342; earlier form 1343)
Labels, machine and growth rate for bounded labels (Theorem 20)	204, 606, 622, 625	checked (630)
Upper bound with α_{b} (Theorem 21)	632	checked (639)
Lower bound with α_{b} (Theorem 21)	643; input 638	checked (650)
Λ algebraic of degree four	625	checked (630)
θ/π and α_{b} irrational	625, 1344	checked (1352, 1355)
α_{b} transcendental	625	proved
Generating function not D-finite (Corollary 22)	1344, 1353	checked (1352, 1355)
Fraction and depth criteria, gluing barrier, correlation condition (Section 7)	review_2026-10-25_ideas_2_3	proved
Existence of α implies a polynomial bound; subexponential reformulation	review_2026-10-25_ideas_2_3	proved
$\beta \approx 1.92$	review_2026-10-25_ideas_2_3	heuristic
Baxter numbers for labels in $\{-1, 0, 1\}$	628	literature, data
Tables 1 and 2, P_n, ratios	this paper; also 628, 120	data

Each step of the proof for bounded labels was read independently once, and none twice. That proof uses the martingale central limit theorem, which was checked at a source, and the series for the Dirichlet heat kernel of a planar wedge, which enters only the lower bound of Theorem 21. That series was at first only tested numerically. It has since been checked against its primary source, Lemma 1 of Bañuelos and Smits [2], and the estimate drawn from it was rederived (1353) and checked independently (1355). The rate in Theorem 3 and the cut bound of Proposition 16, which uses it, rest as stated on the earlier lower bound with $\sqrt{n}\log n$; their improved forms use Theorem 1 instead.

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